

Data Analysis from Genes to Ecosystems

Detailed Program and Session Plan — Version 3, schedule-locked edition 2026/27

Master in Molecular and Environmental Computational Biology (MBCMA) — University of Minho

1. CU identity

Format: Intensive 4-week block delivered across 11 contact days (Wed–Fri or Wed–Thu) in September–October 2026. Daily blocks are typically 3–4 h (morning 9–12/13 h + afternoon 14–17/18 h). Total contact: ~63 h (27 h T + 34 h TP + 2 h written exam).

Edition dates (2026/27):

Week	Dates	Days	T	TP	Exam
W1	Wed 16 – Fri 18 Sep 2026	3	11 h	7 h	—
W2	Wed 23 – Thu 24 Sep 2026	2	6 h	8 h	—
W3	Wed 30 Sep – Fri 2 Oct 2026	3	6 h	8 h	—
W4	Wed 7 – Fri 9 Oct 2026	3	4 h	11 h	2 h
Total		11	27 h	34 h	2 h

Position in the curriculum: First CU of the master's, Semester 1. Functions as an orientation course preceding domain-specific CUs that will treat individual topics in depth (Fundamentals in Computational Biology, Machine Learning in Biosciences, Omics and Genetic Approaches in Model Organisms, Ecoinformatics, Molecular Modelling and Image Analysis).

Audience: MSc students with biology BSc background and limited prior exposure to coding and statistics.

Languages/tools backbone: R (**tidyverse**, **vegan**, **phyloseq**, **ape/ggtree**) is the anchor environment, introduced in Week 1 and reused throughout the master's. Python (Biopython) is included as a light exposure in Week 2. RStudio and Jupyter are the standard environments. AI-guided tools are threaded through TPs as short, critically-framed exercises (W1–W3); a final cross-module AI-reflection note feeds into Q6 in W4.

Lifecycle framing. DABDGE is organised around a clean lifecycle arc that maps natural content shape onto lecturer expertise:

Generation & deposit (W1) → Architecture & FAIR (W2) → Analysis & inference (W3) → Application & integration (W4).

The module should help students understand that biological data are not just numbers, but records tied to samples, metadata, methods, and biological meaning. The arc moves from data generation to data architecture to basic analysis to integrated application, with examples spanning sequence data, gene expression, proteins, phenotypes, and ecological observations.

Lecturer team:

- Week 1: Pedro (UMinho) — full week, with Giorgio as a 2 h guest on Fri morning (ecosystem-scale data acquisition & deposit).
- Week 2: Rutger Vos (Naturalis Biodiversity Center) — full 2-day block.
- Week 3: Pedro — full week.
- Week 4: Filipe Costa (UMinho) — full week including TP supervision and exam invigilation. Pedro authors the exam but is not present in W4 contact hours.

2. Pedagogical philosophy

Five principles guide the design and should be made explicit to students in the opening session:

- **Map-making over mastery.** By the end of the CU each student should be able to place any biological dataset on a mental map: which level of organization it belongs to, which data type it represents, which family of analyses applies, and what kinds of questions it can answer.
- **Shared vocabulary.** Terms used loosely across the master's (FASTQ, ASV, OTU, ordination, FAIR, reproducibility, multiple testing, alpha/beta diversity, effect size, BIN, BAGS) are defined here once and consistently.
- **Critical data literacy.** How to read a figure, judge a sample size, recognize a confounded design, and identify when a claim outruns the data. This is the durable competence that survives changes in tools and platforms.
- **Tool tasting, not tool mastery.** Each TP exposes students to representative tools and workflows without expecting fluency. Depth in any single tool family is delegated to downstream CUs (see §7 hand-off table). Whenever a topic is kept light, the consolidating downstream CU is named explicitly to students.
- **Master's-level scaffolding.** DABDGE is CU1 of a coordinated sequence; the R + RStudio environment introduced here is the one students will reuse across the whole master's. R Markdown / Quarto packaging is a transferable skill from day 2 onwards.

AI-guided tools as a meta-skill. TPs in W1–W3 include a brief AI exercise (typically 15–25 min). Each such exercise is paired with a critical-evaluation prompt: students must identify at least one error, hallucination, oversimplification or methodological limitation in the AI output. In W4 the thread is preserved as a *cross-module AI-reflection note* submitted alongside Q6 (no TP-level AI exercise during Filipe's blocks). The objective is to cultivate epistemic hygiene from the first CU of the programme.

3. Learning outcomes

On completion of this CU, students should be able to:

- Recognize the principal types of biological data, associate each with the appropriate level of biological organization, and identify where each is deposited and retrieved (NCBI/ENA, GBIF, UniProt, PRIDE, BOLD, MGnify).
- Describe core principles of experimental and sampling design, identify common sources of bias and confounding, and judge whether a design supports the inferences drawn from it.
- Read, retrieve, and assess the metadata quality of biological records across multiple public databases; describe FAIR principles and standard ontologies (GO, ENVO, Darwin Core).

- Perform basic exploratory analyses on biological datasets in R (tidyverse) — read data, summarize, visualize, run elementary statistical tests — interpreting outputs in biological terms.
- Carry out a light hands-on analytical workflow across multiple data types (genomics annotation, metabarcoding with phyloseq, α/β diversity and ordination, a small proteomics summary, basic phylogeny).
- Articulate the alignment → similarity → function concept chain at an introductory level (depth handed off to CU2).
- Use AI-guided tools (LLM assistants, specialized scientific AI) productively while identifying their limitations, errors and failure modes.
- Discuss primary literature in integrative biology, articulating the data, analyses, conclusions, and limitations of a paper.
- Produce reproducible analyses with appropriate documentation (R Markdown / Quarto) and follow basic FAIR principles.

4. Overall structure

The CU is organized in four weekly modules, each combining theoretical orientation (T) and hands-on consolidation (TP). The modules follow the lifecycle arc and culminate in an integrative case study and assessment.

Week	Module	Dates	T	TP	Anchor question
1	Foundations + data generation & deposit	16–18 Sep 2026	11 h	7 h	Where does biological data come from, and where does it go?
2	Data architecture, FAIR, ontologies & retrieval	23–24 Sep 2026	6 h	8 h	How is a biological record structured, made interoperable, and used?
3	Hands-on analysis & biological inference	30 Sep – 2 Oct	6 h	8 h	How do we extract biological meaning from these data?
4	BOLD-anchored applications, community → ecosystem, integration, exam	7–9 Oct 2026	4 h	11 h + 2 h exam	How does this all come together in a real biodiversity / ecosystem question?
Total			27 h	34 h + 2 h exam = 63 h	

Daily blocks are typically 3–4 h (morning 9–12/13 h + afternoon 14–17/18 h). Within a day, T and TP segments are interleaved where possible to allow immediate consolidation. The session-level tables below show the daily plan.

Recurring case-study anchor. A coastal/marine system aligned with U. Minho regional observatories serves as the recurring biological anchor. The same site is revisited each week through a different data lens: existing records (W1) → retrieved & integrated metadata (W2) → analyses of one or two

layers (W3) → BOLD curation + integrative report (W4). Assembly of the anchor pool is a parallel work-stream; the program runs without it if needed, with the loss of some narrative cohesion.

5. Assessment

Continuous TP quizzes / mini-reports (60%) + individual written exam (40%). The intensive format makes longer reports impractical; short structured quizzes / mini-reports capture engagement and consolidate the day's key ideas without saturating students, while the exam ensures individual accountability for the conceptual material.

5.1 Continuous TP quizzes / mini-reports (60%)

Across the four weeks, students submit six short quizzes / mini-reports (10% each), one per major TP block. Each consists of 2–4 targeted questions on the day's TP plus the key figure or table the student produced, submitted by end of day. Expected effort: 30–45 minutes to finalise, not a full evening. Grading is rapid (correct / partial / incorrect on each item) and feedback is returned the following morning, so misunderstandings are caught early. Submissions are made as concise reproducible documents (R Markdown / Quarto) when code is involved, or as a single PDF / Word page when only interpretation is required.

ID	Submit by	Quiz / mini-report content
Q1	EOD Fri 18 Sep	Q1: identify two design flaws in a short published methods excerpt. Q2: compute mean, SD and one diagnostic plot on a provided dataset. Q3: state a testable hypothesis. Submit half-page + one figure (RMarkdown/Quarto).
Q2	EOD Thu 24 Sep	Metadata-quality mini-report. Q1: retrieve one biological record from a named public database; report its identifiers, provenance and missing fields. Q2: identify two metadata-quality issues and propose one fix per issue. Submit half-page + a brief annotated record screenshot.
Q3	EOD Fri 25 Sep	Cross-database integration mini-report. Q1: link one sample/specimen across two or more public resources (e.g., GenBank + GBIF, or BOLD + WoRMS); list discrepancies. Q2: describe one FAIR principle the linked record satisfies and one it fails. Submit half-page + one integration diagram or table. (Submitted after W2; finalised over the weekend.)
Q4	EOD Thu 1 Oct	Q1: from a small ASV table (phyloseq), name the dominant taxonomic group and one assumption made by the reference database. Q2: produce one diagnostic figure. Submit half-page + one figure.
Q5	EOD Fri 2 Oct	Q1: report Shannon diversity for two sites and state which is more diverse. Q2: produce one ordination plot (PCA, PCoA or NMDS). Q3: read a small phylogenetic tree and answer two questions about topology and support. Submit half-page + two figures.
Q6	EOD Thu 8 Oct	Integrative multi-scale mini-report on the recurring anchor site. Q1: identify which variables represent which level of organization. Q2: produce one figure linking two scales (e.g., barcode-derived community composition vs an environmental variable). Q3: in three sentences, state the integrative claim the data can support and one it cannot. Plus: cross-module AI-reflection note (one paragraph) on how AI tools were used across W1–W4 and one concrete error / limitation observed in each module. Submit one-page + one figure + reflection.

Grading (each quiz / mini-report): each question scored correct / partial / incorrect; figure scored on technical correctness and readability; brief AI-use note included where the TP used AI tools. Total per quiz: 10 points. Feedback returned the next morning where possible.

5.2 Individual written exam (40%)

Two-hour written exam on Fri 9 Oct 2026. Authored by Pedro; supervised by Filipe in W4 contact hours. Structure:

- Short-answer conceptual questions covering the four modules (50% of exam grade).
- Figure and table interpretation: students are given unfamiliar figures from primary literature and asked to describe the analysis, the message, and one limitation (30%).
- Mini-case: a short scenario with sampling design, hypothesis, and choice of analytical approach to justify (20%).

No code is asked to be written from memory; the exam tests understanding, not syntax.

5.3 AI-use policy

Students may use AI-guided tools (LLM assistants, scientific AI) for any quiz / mini-report, provided that:

- They disclose use in a brief note at the end of each submission (which tool, for what task).
- Where AI was used substantively, they include at least one explicit observation of an AI output they had to correct, reject or refine.

The objective is normalising responsible, transparent use rather than discouraging it. In the exam, AI tools are not available; this is by design — the exam evaluates internalised understanding.

6. Week-by-week detailed program

Week 1 — Foundations + data generation and deposit (Pedro + Giorgio guest)

Dates: Wed 16 – Fri 18 Sep 2026 (3 days; 11 h T + 7 h TP).

Anchor question: *Where does biological data come from, and where does it go?*

Learning objectives

- Distinguish data types (continuous, count, categorical, sequence, image, spatial, temporal) and recognize their domain provenance.
- Describe at an orientation level how the major molecular data types are generated (sequencing, mass spectrometry, microarrays) and what file formats they produce.
- Describe at an orientation level how ecosystem-scale data are acquired (satellite imagery, environmental sensors, citizen science, biogeochemical proxies) and where they are deposited.
- Articulate principles of experimental and sampling design and identify pitfalls (pseudoreplication, confounding, batch effects).
- Apply descriptive statistics and basic inferential tests to small biological datasets.
- Operate minimally in RStudio: open a project, read a CSV, produce a plot, package work as R Markdown / Quarto.
- State and apply FAIR principles and reproducibility practices at an introductory level (deepened in W2).

Theoretical content (T, 11 h)

- What is biological data? Sources, scales, formats. The role of metadata. Levels of biological organization as a framework for analytical choice.
- Generation of molecular data: sequencing technologies (short read, long read, single-cell) in one slide each; mass spectrometry in one slide; microarrays in one slide. File formats produced (FASTA, FASTQ, GenBank, mzML — orientation only).
- Experimental and observational designs. Replication, randomization, blocking. Pseudoreplication and confounding. Sampling design: representativeness, sampling effort, rarefaction.
- Descriptive statistics; distributions; visualization as a diagnostic tool.
- Hypothesis testing essentials: null/alternative, p-values, effect size, multiple testing. Common misinterpretations.
- Public databases and deposit: NCBI/ENA, GBIF, UniProt, PDB, MGnify, PRIDE (used Thu-W3), BOLD (preview for W4). What goes where, and why. Bridges to W2 (architecture / retrieval) and W4 (BOLD).
- Reproducibility, FAIR data, open science (introductory pass). Version control and the literate-programming idea (R Markdown / Quarto).
- Brief introduction to AI-guided tools in scientific data analysis: what they do well, what they fail at, how to evaluate output.

Guest contribution (2 h, Fri 18 Sep, 9–11 h)

- **Giorgio — Ecosystem-scale data acquisition & deposit:** satellite imagery (Copernicus, etc.), environmental sensors, citizen science (GBIF citizen-science layers, OBIS for marine), biogeochemical proxies. What these data look like and what gets deposited where. Frames W4's applied use of these data.

TP content (7 h)

- **TP1.1 — R/RStudio bootstrap + dataset inspection** (Thu 14–17 h, 3 h). RStudio as the master's working environment; first plot from a CSV; R Markdown / Quarto packaging introduced so packaging is a transferable skill from day 2 onwards. Light Python touchpoint (~5 min Jupyter sighting). Then: three biological records (a GenBank entry, a GBIF occurrence, a small phenotype table) — students identify what was measured, how, and what's missing. *If the recurring anchor is available, records are from the anchor site.*
- **TP1.2 — Design critique + EDA + hypothesis-test demo** (Fri 11–13 h, 2 h). Short worked examples of flawed published designs; descriptive/EDA on a small biological dataset (e.g., morphometrics, growth curves, abundance counts); hypothesis-test demo (t-test, non-parametric alternative, multiple-testing correction). AI exercise: critique an LLM-generated methods summary.

Daily schedule (Week 1)

Date	Time	Block	Lecturer	Content
Wed 16 Sep	10–13 h	T (3 h)	Pedro	Welcome; what is biological data; levels of organisation; metadata. Generation of molecular data: sequencing, MS, microarrays (one slide each); file formats produced.
Wed 16 Sep	14–18 h	T (4 h)	Pedro	Experimental & sampling design; descriptive statistics; visualization as diagnostic; hypothesis

Date	Time	Block	Lecturer	Content
				testing essentials; intro to AI tools as research instruments.
Thu 17 Sep	9–13 h	T (4 h)	Pedro	Public databases and deposit (NCBI/ENA, GBIF, UniProt, PDB, MGnify, PRIDE, BOLD preview); Reproducibility & FAIR (introductory pass); further AI-as-instrument framing.
Thu 17 Sep	14–17 h	TP (3 h)	Pedro	TP1.1 R/RStudio bootstrap + R Markdown / Quarto + dataset inspection (3 record types).
Fri 18 Sep	9–11 h	Guest TP (2 h)	Giorgio	Ecosystem-scale data acquisition & deposit: satellite, sensors, citizen science, biogeochemical proxies. Frames W4.
Fri 18 Sep	11–13 h	TP (2 h)	Pedro	TP1.2 Design critique + EDA + hypothesis-test demo. AI exercise: LLM methods-summary critique.
EOD Fri	—	—	—	Submit Q1.

Week 2 — Data architecture, FAIR, ontologies & retrieval (Rutger Vos)

Dates: Wed 23 – Thu 24 Sep 2026 (2 days; 6 h T + 8 h TP).

Anchor question: *How is a biological record structured, made interoperable, and used?*

Learning objectives

- Describe the logic of biological datasets: rows, columns, identifiers, sample vs feature tables, relational links.
- Articulate the central dogma as a data pipeline and recognize the file formats that carry each step (FASTA, FASTQ, GenBank).
- Retrieve biological records programmatically from major public databases and inspect their provenance and metadata.
- State and apply FAIR principles in depth; recognize and use standard ontologies (GO, ENVO, Darwin Core) and taxonomy identifiers.
- Audit metadata quality and integrate records across multiple sources, identifying inconsistencies and discrepancies.

Theoretical content (T, 6 h)

- Data architecture: rows/columns/identifiers; sample sheets vs feature tables; relational thinking. The central dogma as a *data pipeline*.
- Public DB structure in depth: NCBI/ENA, UniProt, GBIF, BOLD; identifiers, accessions, programmatic access (APIs).
- Metadata, ontologies, controlled vocabularies: GO, ENVO, Darwin Core, taxonomy identifiers. Why annotation quality bounds inference.
- FAIR principles in depth; database curation; QA/QC concepts (precursor to Filipe's W4 BOLD curation block).
- Multi-source integration; what cross-omics / cross-scale interpretation can and cannot claim.

TP content (8 h)

- **TP2.1 — BioPython sequence-handling micro-TP** (Thu morning, ~1.5 h). FASTA in/out, basic stats, translation; BLAST-style similarity intuition with a toy database.
- **TP2.2 — Retrieval TP** (Thu morning, ~2 h). Pull a gene record and a species-occurrence record from public DBs (anchor site if available); inspect provenance and metadata. **Submit Q2** (metadata-quality mini-report).
- **TP2.3 — Annotation TP** (Thu afternoon, ~1.5 h). Annotate a small dataset using standard vocabularies (GO terms, Darwin Core fields). AI exercise: critique LLM-generated metadata fields.
- **TP2.4 — Cross-DB integration TP** (Thu afternoon, ~2 h). Link one sample across multiple resources; spot inconsistencies. **Q3** drafted in class; finalised over weekend, submitted EOD Fri 25 Sep.
- **TP2.5 — Mini-integration wrap** (Thu afternoon, ~1 h). Short multi-table biological example (anchor site if available); demonstrate FAIR-compliant packaging.

AI exercises (threaded)

- LLM-generated metadata fields critique: ask an LLM to populate metadata for a record; identify omissions and unsupported entries.
- LLM cross-source disambiguation: ask an LLM to reconcile two conflicting records; verify whether its resolution respects the underlying evidence.

Daily schedule (Week 2)

Date	Time	Block	Lecturer	Content
Wed 23 Sep	9–12 h	T (3 h)	Rutger	Data architecture; rows/columns/identifiers; central dogma as data pipeline. Public DB structure (part 1): NCBI/ENA, UniProt — identifiers, accessions, APIs.
Wed 23 Sep	14–17 h	T (3 h)	Rutger	Public DB structure (part 2): GBIF, BOLD. Metadata, ontologies, Darwin Core, GO, ENVO. FAIR in depth; curation/QA-QC concepts; multi-source integration.
Thu 24 Sep	9–13 h	TP (4 h)	Rutger	TP2.1 BioPython sequence-handling + BLAST-style intuition. TP2.2 Retrieval TP: gene + occurrence records; inspect metadata. Submit Q2 (EOD Thu).
Thu 24 Sep	14–18 h	TP (4 h)	Rutger	TP2.3 Annotation TP with standard vocabularies (AI exercise: metadata-fields critique). TP2.4 Cross-DB integration TP (AI exercise: cross-source disambiguation). TP2.5 Mini-integration wrap.
EOD Fri 25 Sep	—	—	—	Submit Q3 (finalised over weekend).

Week 3 — Hands-on analysis & biological inference (Pedro)

Dates: Wed 30 Sep – Fri 2 Oct 2026 (3 days; 6 h T + 8 h TP).

Anchor question: How do we extract biological meaning from these data?

Stack discipline. Anchor is **R-tidyverse + phyloseq**. Thu opens with a brief R/RStudio refresh (~10–15 min) before the genomics TP; the same stack carries through metabarcoding (Thu–Fri AM),

proteomics (Fri AM), and phylogeny (Fri AM). Proteomics-Fri reads a flat PRIDE summary table — still R-tidyverse, no DE pipeline. Phylogeny uses an R tree-reading workflow (ape / ggtree).

Learning objectives

- Articulate the alignment → similarity → function concept chain (depth handed off to CU2).
- Read an annotated genome subset and use BLAST-style similarity intuition to infer function for a few ORFs.
- Outline the metabarcoding workflow and the role of reference databases; carry out a phyloseq workflow from a precomputed ASV table to taxonomic summary, α/β diversity, and ordination.
- Distinguish shotgun metagenomics from amplicon metabarcoding conceptually.
- Inspect a small proteomics summary table from PRIDE and produce basic exploratory plots.
- Read a published phylogenetic tree (topology, support, common misreadings); build and interpret a simple tree in R.

Theoretical content (T, 6 h, all Wed)

- From similarity to function (light): closing the W2 thread. Genomics conceptually — from reads to assembly to annotation; what an annotated gene record actually claims.
- Metabarcoding workflow: marker genes, OTU vs ASV, reference databases (precursor to Filipe W4).
- Metagenomics: shotgun vs amplicon; functional profiling intuition; α/β diversity concepts; ordination overview (PCA, PCoA, NMDS).
- Proteomics: mass-spec data structure, peptide-spectrum match concept (light); what a proteomics data table looks like.
- Phylogeny basics: trees as data, support, reading without misreading; brief glance at NJ / ML / Bayesian (one slide each — no algorithm derivation, that goes to CU2).

TP content (8 h)

- **TP3.1 — R/RStudio refresh + genomics hands-on** (Thu morning, ~2 h). Explore a small annotated genome subset (anchor-site organism if available); BLAST-style similarity to infer function for a few ORFs. AI exercise: critique an LLM annotation.
- **TP3.2 — Phyloseq metabarcoding** (Thu morning, ~2 h). From a precomputed ASV table to a taxonomic summary; discuss reference-DB assumptions. **Submit Q4.**
- **TP3.3 — α/β diversity + ordination** (Fri morning, ~1.5 h). Continuing in phyloseq on the same dataset (cumulative workflow).
- **TP3.4 — PRIDE proteomics exploration** (Fri morning, ~1 h). Load a flat summary table in R; basic exploratory plots. AI exercise: critique LLM interpretation of an abundance result. *No DE pipeline — that's CU4 territory.*
- **TP3.5 — Phylogeny** (Fri morning, ~1.5 h). Align a small set of sequences (ideally from the anchor site), build a quick tree in R (ape / ggtree), interpret. **Submit Q5** (combined diversity + tree-reading mini-report).

AI exercises (threaded)

- Ask an LLM to interpret an unfamiliar annotation; check whether it overclaims function from similarity alone.
- Ask an LLM to recommend a diversity index for a scenario; evaluate whether the recommendation is justified or generic.

- Ask an LLM to interpret a proteomics abundance result; identify any overinterpretation or unsupported causal language.

Daily schedule (Week 3)

Date	Time	Block	Lecturer	Content
Wed 30 Sep	9–12 h	T (3 h)	Pedro	From similarity to function (light); genomics conceptually — reads to annotation. Metabarcoding workflow; OTU vs ASV; reference databases (precursor to Filipe W4).
Wed 30 Sep	14–17 h	T (3 h)	Pedro	Metagenomics: shotgun vs amplicon; α/β diversity; ordination overview. Proteomics: MS data structure; PSM (light). Phylogeny basics: trees as data, support; NJ / ML / Bayesian (one slide each).
Thu 1 Oct	9–13 h	TP (4 h)	Pedro	TP3.1 R/RStudio refresh + genomics hands-on (AI exercise: LLM annotation critique). TP3.2 Phyloseq metabarcoding. Submit Q4 (EOD Thu).
Fri 2 Oct	9–13 h	TP (4 h)	Pedro	TP3.3 α/β diversity + ordination (phyloseq, same dataset). TP3.4 PRIDE proteomics exploration (AI exercise: LLM abundance critique). TP3.5 Align + build quick tree in R. Submit Q5 (EOD Fri).

Week 4 — BOLD-anchored applications, community → ecosystem, integration, exam (Filipe Costa)

Dates: Wed 7 – Fri 9 Oct 2026 (3 days; 4 h T + 11 h TP + 2 h exam).

Anchor question: How does this all come together in a real biodiversity / ecosystem question?

Lecturer arrangement. Filipe Costa runs the full week alone (TPs, integrative workshop, and Friday exam supervision). Pedro authors the written exam but is not present in W4 contact hours. The AI thread of W1–W3 is preserved in W4 as a *cross-module AI-reflection note* embedded in Q6 — students reflect on AI use across the four weeks as part of the Q6 submission (no in-class AI exercise during Filipe's blocks).

Learning objectives

- Describe DNA barcoding as a concept and BOLD as a biodiversity-informatics platform; navigate BOLD's structure (workbench vs database, specimen-taxonomy-genetic linkages).
- Use BINs and the BARCODE ID and data portal at an introductory level; retrieve and inspect data.
- Apply curation principles (BAGS, BOLDcurator) to a small mined dataset; cross-reference with WoRMS.
- Connect reference-database-anchored biodiversity assessments to community / ecosystem-scale questions (eDNA monitoring, biodiversity indices, environmental management).
- Construct a simple multi-scale narrative that links molecular, community, and ecosystem evidence around a single biological question on the anchor site.
- Critically read and discuss an integrative biology paper, identifying its data architecture, claims, and limitations.

Theoretical content (T, 4 h, Wed AM)

- DNA barcoding: concept and relevance.
- BOLD's role, placement and interoperability within the biodiversity-data ecosystem.
- BOLD content and structure: workbench (private) vs database (public) including mined GenBank data; specimen + taxonomy + geographic data linked with genetic-marker data; main metadata fields (Darwin Core, GPS, image library, etc.) vs genetic fields (sequences, primers, traces); annotation flags. Projects vs Datasets — features and applications.
- BINs — concept and application.
- Main BOLD analysis tools: taxon ID trees, distance summary, BIN discordance reports.
- Curation & annotation (QA/QC): sources of error (biological and operational); curation guidelines (contrasting morphological and genetic/MOTU data; metadata relevance); BAGS rationale and application; BOLDcurator introduction.
- Bridge to community → ecosystem: how reference-database-anchored biodiversity assessments link to ecosystem signals (eDNA monitoring, biodiversity indices in environmental management). Uses Giorgio's W1-Fri introduction as background context.

TP content (11 h)

- **TP4.1 — BOLD tutorial** (Wed 14–17 h, 3 h). Explore BOLD content; usage of BARCODE ID; data retrieval; small practical exercises.
- **TP4.2 — Curation group exercise** (Thu 9–13 h, 4 h). Mine BOLD on the **anchor-site barcode records**, align + simple tree in MEGA, cross-reference with WoRMS, apply BAGS principles.
- **TP4.3 — Integrative case study workshop + paper reading + figure drills** (Thu 14–18 h, 4 h). Build the multi-scale narrative on the anchor site, linking W1 records → W2 retrieved metadata → W3 analyses → W4 BOLD curation. Guided critical reading of one integrative paper (anatomy of the integration; claims and limitations). Short figure-interpretation drills on unfamiliar published figures (exam-prep folded in). Draft and submit **Q6** EOD Thu, including the cross-module AI-reflection note.

Daily schedule (Week 4)

Date	Time	Block	Lecturer	Content
Wed 7 Oct	9–13 h	T (4 h)	Filipe	DNA barcoding & BOLD: structure (workbench vs database; specimen + taxonomy + genetic), main fields, annotation flags; Projects vs Datasets; BINs; main analysis tools. Curation theory: sources of error, BAGS, BOLDcurator. Bridge to community → ecosystem (eDNA, indices, environmental management).
Wed 7 Oct	14–17 h	TP (3 h)	Filipe	TP4.1 BOLD tutorial: explore content; BARCODE ID; data retrieval; small practical exercises.
Thu 8 Oct	9–13 h	TP (4 h)	Filipe	TP4.2 Curation group exercise — mine BOLD on anchor-site records; align + simple tree in MEGA; cross-reference WoRMS; BAGS principles.
Thu 8 Oct	14–18 h	TP (4 h)	Filipe	TP4.3 Integrative case study workshop + paper reading + figure-interpretation drills (exam-prep folded in). Submit Q6 EOD Thu (incl. cross-module AI-reflection note).

Date	Time	Block	Lecturer	Content
Fri 9 Oct	(2 h slot)	Exam	Filipe (sup.)	Written exam (2 h). Pedro & Filipe

7. Continuing threads and downstream hand-offs

Several threads run across the four modules and should be reinforced verbally in each week:

- **From observation to evidence.** Every dataset is the product of choices about what to measure, how to sample, and how to record. Read those choices first.
- **Visualisation as diagnosis.** Plots are not decoration; they are the first instrument of quality control.
- **Reference frames matter.** Whether the reference is a database, a model organism, a phylogeny, or a baseline period, the choice constrains the conclusions.
- **Reproducibility is a competence, not a virtue.** Practiced from Week 1, demanded in every deliverable thereafter.
- **AI is an instrument with calibration drift.** Useful, often impressive, occasionally wrong in confident ways. Treat its output as you would any instrument output: with documented verification.
- **Scale-linking as the master skill.** DABDGE is the *only* CU in the master's that explicitly works the molecular → community → ecosystem connection. Reinforce it every week.

Downstream hand-offs (what DABDGE deliberately keeps light)

Each downstream CU consolidates a specific competence. DABDGE orients and exposes; the listed CU goes deep.

Topic in DABDGE	Consolidating CU
Alignment algorithms (dynamic programming, scoring matrices)	CU2 <i>Fundamentals in Computational Biology</i>
Phylogenetic inference methods (model selection, MCMC, model comparison)	CU2 / Project
Machine learning (any depth)	CU3 <i>Machine Learning in Biosciences</i>
RNA-seq DE pipelines, full omics workflows	CU4 <i>Omics and Genetic Approaches in Model Organisms</i>
Community-ecology modelling beyond diversity & ordination	CU5 <i>Ecoinformatics</i>
Protein structure prediction (AlphaFold / ESMFold), cellular imaging, flow cytometry, bioenergetics	CU6 <i>Molecular Modelling and Image Analysis</i>
Networks / systems biology	downstream / Project

8. Indicative resources

These are starting points; instructors should refresh per edition.

Books and core references

- Holmes & Huber. Modern Statistics for Modern Biology (open access).
- Wickham & Grolemund. R for Data Science (open access).
- Buffalo. Bioinformatics Data Skills, O'Reilly.
- Hugo et al. The Biostar Handbook (online).
- Borcard, Gillet & Legendre. Numerical Ecology with R.

Tools and platforms

- **Anchor stack (R):** tidyverse, vegan, phyloseq, ape, ggtree, ggplot2.
- **Light Python exposure (W2 under Rutgers):** Biopython.
- **Reproducibility:** RStudio + R Markdown / Quarto; Git/GitHub (light introduction).
- **W4 / barcoding-specific:** BOLD, BOLDcurator, MEGA, WoRMS.
- **AI-guided:** SciMentor AI; general LLM assistants; with critical-evaluation framing in W1–W3 TPs and a cross-module reflection in Q6.

Public databases used across the CU

- **Sequences and genomes:** NCBI / ENA, GenBank.
- **Proteins:** UniProt.
- **Proteomics:** PRIDE (W3-Fri light dataset).
- **Biodiversity records:** GBIF, OBIS (marine), iNaturalist.
- **Microbiome:** MGnify, Earth Microbiome Project subsets.
- **Barcoding:** BOLD (W4 focus); WoRMS as cross-reference.

Teaching datasets (illustrative pool)

- The recurring case-study anchor (coastal/marine system, parallel work-stream) — preferred when available.
- Public 16S amplicon datasets (e.g., Earth Microbiome Project subsets, MGnify studies).
- A small annotated genome subset for the W3-Thu genomics TP.
- A small PRIDE subset for W3-Fri.
- iNaturalist / GBIF subsets for biodiversity exercises.

9. Implementation notes

- **Long daily blocks.** Days run 3–4 h continuous in morning and afternoon. Protect a 10–15 min consolidation break in the middle of each long block. Avoid introducing new tools in the last hour of any block.
- **W4 single-lecturer ownership.** Filipe Costa runs all W4 contact hours including the integrative workshop and exam supervision. Coordinate the integrative workshop scope with Filipe in advance so he can frame Q6 around the BOLD-anchored case study.

- **R bootstrap timing.** The R/RStudio bootstrap and R Markdown / Quarto packaging skill are introduced in Thu-W1 (14–17 h). Q1 finalisation Friday EOD assumes Thursday's bootstrap landed; protect that session.
- **Giorgio guest slot.** Confirm Giorgio for Fri 18 Sep 9–11 h. Scope: ecosystem-scale data acquisition & deposit (satellite, sensors, citizen science, biogeochemical proxies). Light, framing-style — bridges to W4.
- **W2 marathon Thursday.** Thu 24 Sep is an 8 h TP day under Rutger. Plan for two clear breaks (mid-morning and mid-afternoon) and interleave short discussion sections inside each TP block.
- **W3 Wed marathon.** Wed 30 Sep is a 6 h T day under Pedro. Use the day to cover the full conceptual map (genomics → metabarcoding → metagenomics → diversity → proteomics → phylogeny). Reserve high-energy slots for the new conceptual pieces (proteomics, phylogeny basics) which are unfamiliar to most students.
- **Anchor dataset.** Identifying and curating the recurring case-study anchor before the edition strengthens the integration arc and unifies Q1–Q6. Doesn't block running the CU but is a high-value parallel work-stream. Suggestion: align with a U. Minho regional coastal observatory.
- **Provide pre-configured RStudio environments** (e.g., container or institutional server) to remove installation friction in W1. Python (Jupyter + Biopython) is provided for the W2 micro-TP under Rutger only.
- **Cross-CU coordination.** Confirm with Björn (CU2 Python owner) the state DABDGE leaves students in for CU2 entry; confirm §7 hand-offs with the CU3 / CU4 / CU5 / CU6 leads.
- **Q3 weekend submission.** Because W2 ends Thursday but Q3 is the cross-DB integration mini-report (substantial), submission is set to EOD Fri 25 Sep. Communicate this clearly to students at the start of W2.
- **Final feedback round** at end of W4 (asynchronous after the exam) on what worked and what to adjust in the next edition — improves the CU iteratively.